C-06: codon use in the supplied reporter reference

How does the reference CDS compare with historical highly expressed yeast codon counts?

Materials and methods

G1: construct_reference.gbk; original project GenBank reference.

R1: yeast_codon_reference.cut; EMBOSS Eyeast_cai, 24 CDS, TranstermMay1994.

Format only LOCUS metadata spacing in memory; verify ORIGIN bases and raw-file hashes are unchanged. Extract the labelled CDS and translate with standard nuclear code 1; record adjacent stop separately.

Derive weights from integer counts within full synonymous families; replace true zeros by 0.5; exclude Met, Trp and stops from CAI. Sensitivity: zero = 1.

Compare whole CDS and first 50 codons; measure GC by position and flag low-weight clusters and A/T homopolymers.

Scope and working assumptions

The supplied GenBank reference is the analysis sequence. Codon weights use integer counts from the historical 24-CDS yeast table, with the stated zero-count convention. This sequence-level comparison covers coding integrity and codon composition; tested-clone identity, strain-specific expression, transcript context and cadmium performance are outside its scope.

Results

The labelled CDS spans reference bases 748-2097: 1350 nt and 450 aa; its translation matches its annotation.

The adjacent triplet is TAA; a start-to-adjacent-stop segment has 1353 nt.

Count-derived yeast-reference CAI is 0.1708, versus 0.1495 in the first 50 codons; zero = 1 gives 0.1764.

There are 185/450 low-weight codons and 40 consecutive stretches of at least 2 codons; GC is 40.37% and GC3 is 35.78%.

Metric	Value	Input data	Finished without more wet-lab data?
Labelled CDS	1350 nt; 450 aa	G1	Yes: reference scope
Reference coordinates	748-2097	G1	Yes: reference scope
Adjacent stop	TAA; 2098-2100	G1	Yes: reference scope
CAI, whole CDS	0.1708	G1 + R1	Yes: reference scope
CAI, first 50 codons	0.1495	G1 + R1	Yes: reference scope
CAI, zero-count replacement 1	0.1764	G1 + R1	Yes: reference scope
GC / GC3	40.37% / 35.78%	G1	Yes: reference scope
Codons with weight < 0.2	185/450	G1 + R1	Yes: reference scope
Consecutive low-weight stretches	40	G1 + R1	Yes: reference scope
Longest A / T homopolymer	5 / 8 nt	G1	Yes: reference scope
N-terminal His stretch	HHHHHH	G1	Yes: reference scope

Interpretation

The 1350-nt reporter CDS encodes 450 amino acids and matches its annotated translation. Its yeast-reference CAI is 0.1708, compared with 0.1495 in the first 50 codons. There are 185 low-weight codon positions and 40 consecutive stretches of at least 2 codons, locating regions for a synonymous-redesign comparison.

References and reproducibility

Sharp & Li (1987), doi:10.1093/nar/15.3.1281.

Xia (2007), doi:10.1177/117693430700300028.

EMBOSS CAI manual: https://emboss.sourceforge.net/apps/cvs/emboss/apps/cai.html

Count source: https://sources.debian.org/data/main/e/emboss/6.6.0%2Bdfsg-15/emboss/data/CODONS/Eyeast_cai.cut

Complete inline code: tasks/C-06/C-06.ipynb. Numerical data and source/software provenance: data/output/C-06/.

Figures

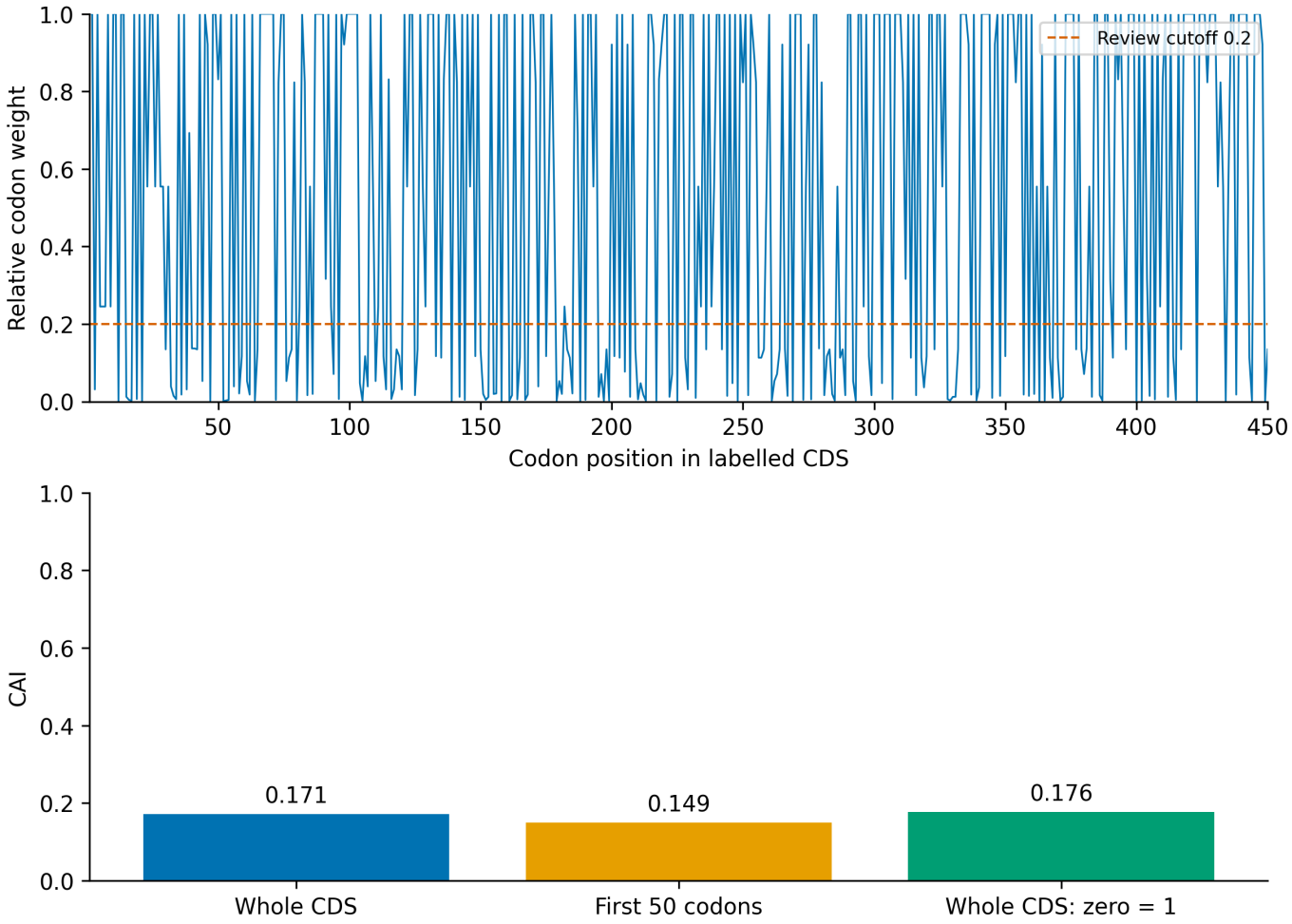


Figure 1. G1 CDS weights and CAI using R1 integer counts. CAI excludes Met/Trp; zero counts replaced by 0.5, with zero = 1 sensitivity. Cutoff 0.2 is descriptive. One reference sequence; no biological error bars.

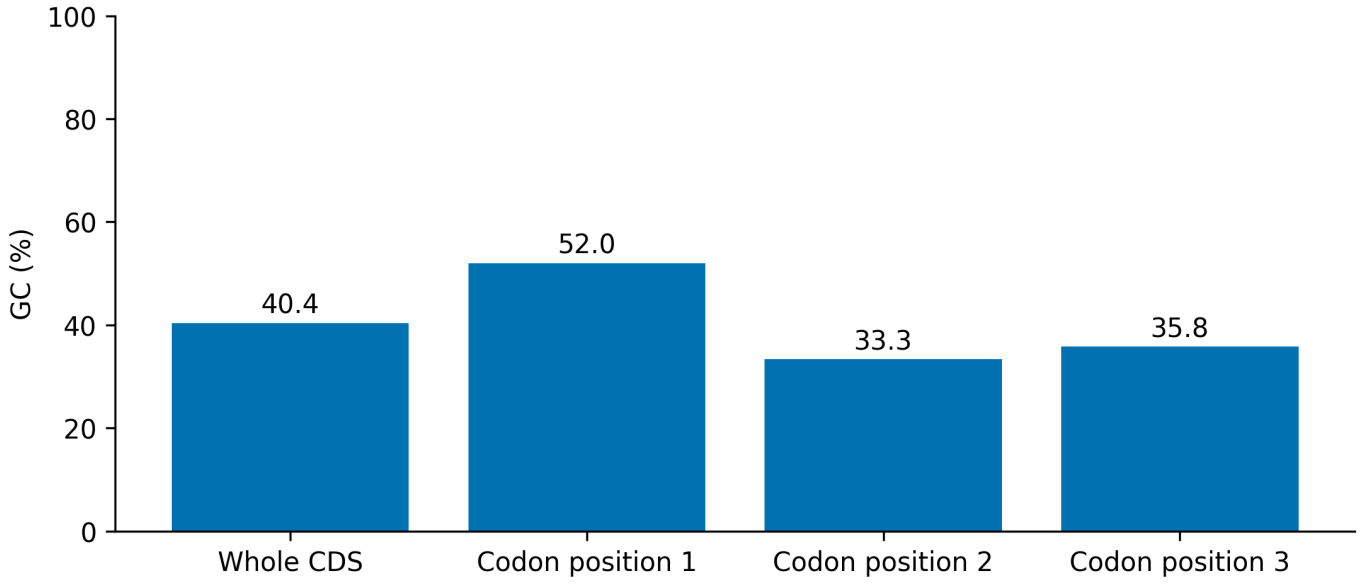


Figure 2. GC percentage in G1 overall and by codon position. One deterministic reference sequence; no biological error bars.